21-3a The Consumer’s Optimal Choices

Once again, consider our pizza and Pepsi example. The consumer would like to end up with the best possible combination of pizza and Pepsi for her—that is, the combination on her highest possible indifference curve. But the consumer must also end up on or below her budget constraint, which measures the total resources available to her.

Figure 7 shows the consumer’s budget constraint and three of her many indifference curves. The highest indifference curve that the consumer can reach (I_2 in the figure) is the one that just barely touches her budget constraint. The point at which this indifference curve and the budget constraint touch is called the *optimum*. The consumer would prefer point A, but she cannot afford that bundle of goods because it lies above her budget constraint. The consumer can afford point B, but that bundle of goods is on a lower indifference curve and, therefore, provides the consumer less satisfaction. The optimum represents the best bundle of pizza and Pepsi that the consumer can afford.

Figure 7

The Consumer’s Optimum

The consumer chooses the point on her budget constraint that lies on the highest indifference curve. At this point, called the optimum, the marginal rate of substitution equals the relative price of the two goods. Here the highest indifference curve the consumer can reach is I_2. The consumer prefers point A, which lies on indifference curve I_3, but she cannot afford this bundle of pizza and Pepsi. By contrast, point B is affordable, but because it lies on a lower indifference curve, the consumer does not prefer it.

Notice that, at the optimum, the slope of the indifference curve equals the slope of the budget constraint. We say that the indifference curve is *tangent* to the budget constraint. The slope of the indifference curve is the marginal rate of substitution between pizza and Pepsi, and the slope of the budget constraint is the relative price of pizza and Pepsi. Thus, *the consumer chooses the quantities of the two goods so that the marginal rate of substitution equals the relative price.*

In Chapter 7, we saw how market prices reflect the marginal value that consumers place on goods. This analysis of consumer choice shows the same result in another way. In making her consumption choices, the consumer takes the relative price of the two goods as given and then chooses an optimum bundle of goods at which her marginal rate of substitution equals this relative price. The relative price is the rate at which the *market* is willing to trade one good for the other, whereas the marginal rate of substitution is the rate at which the *consumer* is willing to trade one good for the other. At the consumer's optimum, her valuation of the two goods (as measured by the marginal rate of substitution) equals the market's valuation (as measured by the relative price). As a result of this consumer optimization, market prices of different goods reflect the value that consumers place on those goods.